curios-fits-imaging README

Created by	Steven V W Beckwith
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Tags	

▼ CuRIOS FITS Imaging

Python tools for reading, analyzing, and visualizing astronomical FITS images acquired with the CuRIOS imaging system.

The package provides a modern workflow for:

- Reading FITS images
- Detecting stellar sources
- Measuring centroids and photometry
- Displaying publication-quality image plots
- Inspecting source cutouts
- Performing basic PSF diagnostics
- Exporting measurements for further analysis

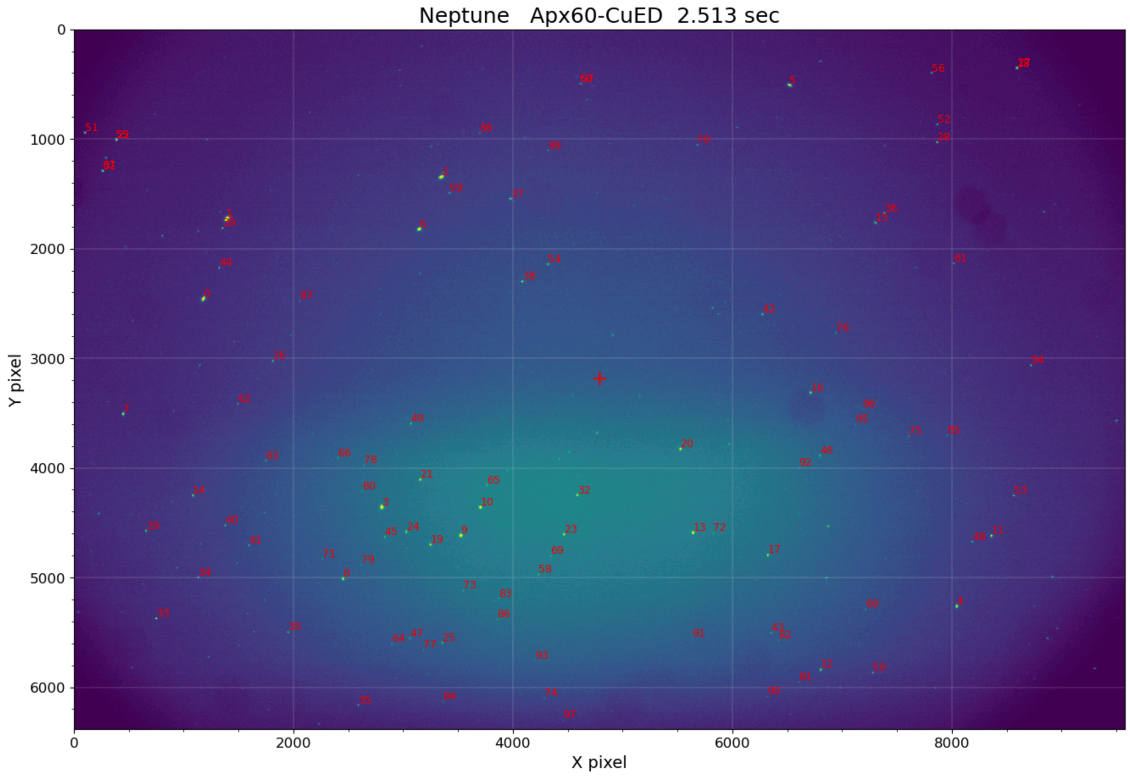
The project is designed to be used from Jupyter Lab today and will evolve into a standalone scientific workbench with both command-line and graphical interfaces.

▼ Example

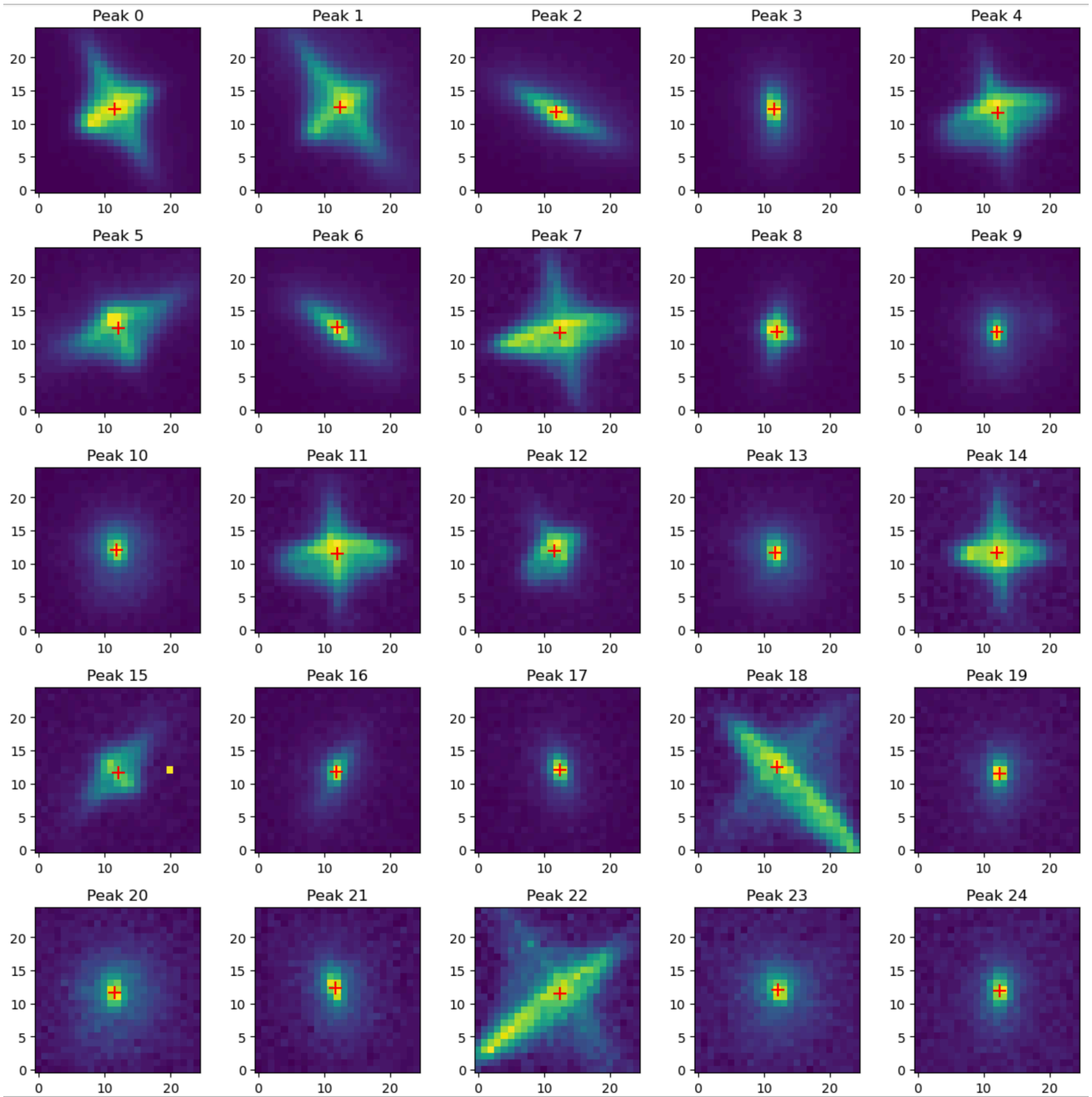
```python
python

from pathlib import Path

from fits_imaging.analysis import analyze_image
from fits_imaging.config import ImagingConfig
from fits_imaging.session import ImagingSession

data_root = Path("/path/to/your/DataDirectories")
config = ImagingConfig(data_root=data_root)
session = ImagingSession(data_root=data_root)

session.folder_table()
session.set_folder_by_index(0)

run = session.open_run()
run.preview(kind="science", n=25)

image_path = session.set_image_by_index(0, kind="scienc
e")
results = analyze_image(image_path, config)
results.report(config=config)
```

Screenshots

```
Neptune  9576 x 6380  Apx60-CuED  2.513 sec
RAH:   0.283  Dec:   0.348  PA: 315.64  PA2:  44.36  Alt:    31.5  Az:   117.8
Mean: 302.5  Median: 301.0  Std: 29.8  Min: 91.0  Max: 37624.0
Found 98 peaks in 2.98 sec

Source statistics:
{'nstars': 98, 'median_fwhm': 5.722994804382324, 'median_flux': 10102.0, 'brightest_flux': 412565.0}

Selected-region source statistics:
{'nstars': 26, 'median_fwhm': 5.5563459396362305, 'median_flux': 12822.0, 'brightest_flux': 155481.0}
```

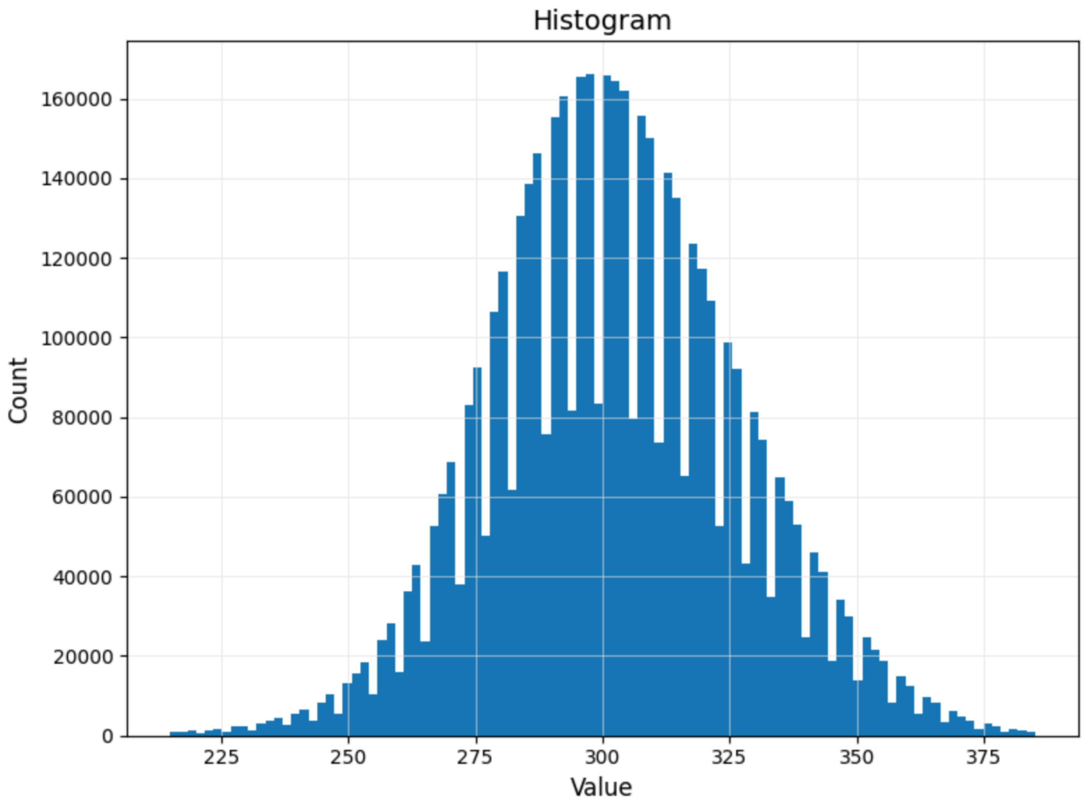


Peak#	X	Y	Area	sx	sy	Flux	Mag
0	1177.55	2458.14	613561.6	3.93	3.46	412565.0	7.113
1	1392.36	1726.46	408838.0	4.29	4.66	221865.0	7.787
2	3345.88	1348.74	223432.6	3.11	2.13	189235.0	7.960
3	2804.54	4357.14	167160.5	1.38	2.54	155481.0	8.173
4	8045.11	5260.66	213413.9	3.72	3.00	151463.0	8.201
5	6522.11	511.26	189935.2	3.53	3.28	138421.0	8.299
6	3144.93	1824.45	147813.7	2.67	2.34	137781.0	8.304
7	447.36	3507.63	172392.4	5.16	3.04	89506.0	8.773
8	2450.95	5010.83	86552.6	1.62	2.08	86442.0	8.810
9	3524.00	4615.77	75465.9	1.81	2.45	77563.0	8.928
10	3702.84	4360.11	73996.7	2.28	2.72	69531.0	9.047
11	8360.88	4618.51	100179.9	3.83	2.94	69472.0	9.048
12	6805.60	5838.96	68690.1	2.43	2.59	62752.0	9.158
13	5641.76	4590.59	61403.2	2.17	2.63	59175.0	9.222
14	1082.03	4250.58	58370.8	3.75	2.47	41454.0	9.608
15	7304.11	1765.62	49133.6	2.63	2.83	40137.0	9.643
16	6713.88	3314.77	33764.6	1.58	2.20	36444.0	9.748
17	6323.36	4794.05	30726.3	1.35	2.00	34446.0	9.809
18	8590.94	358.43	54838.1	4.13	4.23	32517.0	9.872
19	3248.39	4699.51	30818.6	1.64	2.36	31950.0	9.891
20	5526.55	3828.57	38914.9	2.88	3.29	31186.0	9.917
21	3154.65	4107.38	31059.2	2.10	3.04	27061.0	10.071
22	388.39	1011.51	47280.3	5.07	4.06	24506.0	10.179
23	4465.16	4606.05	27006.1	2.59	2.82	24337.0	10.187
24	3028.45	4582.83	21495.7	1.82	2.59	20943.0	10.350
25	3356.26	5594.37	18530.6	1.43	1.82	20247.0	10.386
26	1812.20	3026.44	21851.3	1.84	2.61	18607.0	10.478
27	8598.50	350.55	23064.9	2.57	2.86	18108.0	10.508
28	7866.68	1032.66	27293.3	3.52	3.53	17725.0	10.531
29	655.23	4573.69	26370.5	4.27	2.99	16927.0	10.581
30	1950.64	5497.31	19315.1	2.81	2.32	16883.0	10.584
31	263.73	1295.73	27667.4	4.46	3.37	16704.0	10.595
32	4586.40	4248.50	21442.5	3.25	3.46	16036.0	10.639
33	748.90	5371.94	32139.0	5.23	3.20	15661.0	10.665
34	1132.54	4998.04	22077.7	3.96	2.60	14790.0	10.727
35	2588.98	6163.63	17077.8	2.34	2.57	14754.0	10.730
36	7386.38	1678.53	17788.4	2.62	2.79	14014.0	10.786
37	3977.26	1547.70	18936.6	3.59	2.09	13884.0	10.796
38	4084.37	2301.99	17428.9	3.41	2.69	13735.0	10.808
39	1353.45	1815.57	21575.5	4.03	3.97	13240.0	10.848
40	1378.04	4523.25	16177.0	3.24	2.55	12772.0	10.887
41	1595.33	4706.03	14585.6	2.61	2.38	12018.0	10.953
42	6273.02	2601.27	12688.7	2.50	2.57	11909.0	10.963
43	6349.92	5502.92	11382.3	1.60	2.05	11683.0	10.983
44	1322.50	2176.11	15554.7	3.51	3.17	11275.0	11.022
45	2832.29	4630.30	11396.2	1.37	2.41	11235.0	11.026
46	6797.85	3888.77	12046.8	1.46	2.56	11224.0	11.027
47	3060.77	5553.33	9986.8	1.59	1.90	10571.0	11.092
48	8186.28	4673.36	14372.9	3.75	2.56	10169.0	11.134
49	3069.35	3599.10	10762.3	2.14	2.80	10035.0	11.148



Peak 0
Peak 1
Peak 2
Peak 3
Peak 4
Peak 5
Peak 6
Peak 7
Peak 8
Peak 9
Peak 10
Peak 11
Peak 12
Peak 13
Peak 14
Peak 15
Peak 16
Peak 17
Peak 18
Peak 19
Peak 20
Peak 21
Peak 22
Peak 23
Peak 24

```
results.plot_histogram(config=config)
plt.show()
```



▼ Features

- FITS image I/O
- Automatic source detection
- Gaussian and moment-based centroid estimation
- Instrumental photometry
- Image statistics
- Region-based statistics
- Image cutouts
- Peak tables
- CSV export
- Radial PSF profiles
- Encircled-energy calculations
- Persistent imaging sessions
- Observing-run file tables
- Image-kind filtering for science, focus, dark, flat, and rotation/blur files
- Command-line batch entry point
- Configurable plotting styles

Repository Layout

```
curios-fits-imaging/
├── fits_imaging/
│   ├── analysis.py
├──│   ├── config.py
│   ├── diagnostics.py
│   ├── export.py
```

```
│   ├── fits_io.py
│   ├── models.py
│   ├── peak_finding.py
│   ├── photometry.py
│   ├── plotting.py
│   ├── plot_style.py
│   ├── psf.py
│   ├── report.py
│   ├── regions.py
│   └── session.py
│
├── notebooks/
├── examples/
├── tests/
├── README.md
└── CHANGELOG.md
```

Installation

Clone the repository

````
```bash

git clone https://github.com/<your-account>/curios-fits-imaging.git
cd curios-fits-imaging
````

Make sure you have Python 3.12 installed

````
```bash
````

````
% brew update
% brew install python3.12      (Mac)

$ sudo apt update
$ sudo apt install python3.12 python3.12-venv python3.12-dev    (Linux)
````

Create a virtual environment

````
```bash

python3.12 -m venv .venv
source .venv/bin/activate
````

Install the scientific packages

````
```bash

python -m pip install -e .
python -m pip install jupyterlab ipykernel
````

This installs the `curios-fits` command-line program and makes the `fits_imaging` package importable from notebooks.

To install the desktop GUI dependencies:

````
```bash

python -m pip install -e ".[gui]"
````

Typical Workflow
````

Jupyter Lab

1. Start Jupyter Lab.
2. Open `notebooks/CuRIOS_Image_Analysis.ipynb`.
3. Set `data_root` to the folder that contains your observing folders.
4. Select an observing folder.
5. Select a FITS image.
6. Run the analysis.
7. Inspect the generated report.
8. Export results if desired.

For folder-level work:

````
```python

run.preview(kind="science", n=25)
run = session.open_run()
summary = run.analyze_all(config, kind="science", max_files=10)
````

`run.file_table` is filename-based by default so it appears quickly even for large or cloud-backed FITS folders. To read FITS headers for object, camera, single-frame exposure, total exposure, and date columns, run:

````
```python

run.load_file_metadata()
````

Summed files with names such as `Target_..._5_sum.fits` are treated as five frames. The original FITS exposure is kept as `exposure_sec`; the derived total appears as `total_exposure_sec`. Photometry normalization uses `photometry_exposure_sec`, which defaults to the total exposure.

Use `kind="science_single"`, `kind="science_sum"`, `kind="focus"`, `kind="dark"`, `kind="flat"`, or `kind="rotation_blur"` to choose a specific image type. Use `kind=None` only when you intentionally want every FITS file in the folder to go through the current source-detection analysis. Analysis modes can also be selected explicitly:

```python

run.analyze_all(config, kind="dark", mode="statistics")
run.analyze_all(config, kind="focus", mode="focus")
run.analyze_all(config, kind="rotation_blur", mode="rotation_blur")
```

If `mode` is omitted, `ImagingRun` chooses a default from each file's kind:

- Science images use `source_detection`
- Focus/Bahtinov images use `focus`
- Rotating or blurred fields use `rotation_blur`
- Darks and flats use `statistics`

Command Line

List the files and their inferred image types:

```bash

curios-fits /path/to/observing-folder --list
```

Include FITS header metadata in the listing:

```bash

```

````bash
curios-fits /path/to/observing-folder --list --metadata
````

Analyze science images in one observing folder:

````
```bash

curios-fits /path/to/observing-folder --kind science --output summary.csv
````

Tune source-detection parameters:

````
```bash

curios-fits /path/to/observing-folder \
--kind science \
--threshold-sigma 7 \
--max-peaks 200 \
--peak-separation 12 \
--fit-method gaussian
````

Run a specific analysis mode:

````
```bash

curios-fits /path/to/observing-folder --kind dark --mode statistics
curios-fits /path/to/observing-folder --kind focus --mode focus
````

The command-line tool defaults to `--kind science` so calibration and focus frames are not accidentally sent through source detection.

Desktop GUI

Launch the first PySide6 workbench:
````

```bash
curios-fits-gui
```

The current GUI milestone supports:

- Selecting one observing folder
- Browsing files by image type
- Previewing a selected FITS image
- Adjusting source-detection parameters
- Running one selected-file analysis
- Displaying a compact result table
- Listing detected peaks after source detection
- Exporting the selected-image summary
- Opening a peak-inspector window with a source cutout, position, flux, magnitude, Gaussian widths, and FWHM

Batch GUI analysis and polished image/plot export are planned later. The GUI uses the same `ImagingConfig`, `ImagingRun`, and `analyze_image` paths as the notebook and CLI.

Development Status

Current release:

- ***Version 2.9***

Implemented

- ImagingResults API
- Session management
- Quick-look report
- Region-aware statistics
- PSF diagnostics

- CSV export
- Observing-run API
- Command-line entry point
- Explicit analysis modes for source detection, focus, rotation/blur, and statistics-only workflows
- Command-line controls for core peak-finding parameters
- First PySide6 desktop workbench

Planned

- WCS support
- More specialized Bahtinov focus measurements
- Fixed-width Gaussian fitting option
- Batch-oriented GUI analysis
- GUI image/plot export
- --

Development Setup

Create the virtual environment

````
```bash

python3.12 -m venv .venv312
source .venv312/bin/activate
````

Install the package in editable mode

````
```bash

pip install -e .
````

Install the Jupyter kernel

````
```bash

python -m ipykernel install \
--user \
--name curios312 \
--display-name "Python 3.12 (CuRIOS)"
````

Launch Jupyter Lab

````
```bash

jupyter lab
````

License

(To be added.)

- --

Author

Steven Beckwith

CuRIOS Imaging Project